

General Education Reflection

By Randolph Roman

Through my time here at Iowa State University, I have taken many classes related to my major in computer engineering. ISU's curriculum offers a diverse range of classes to expand your knowledge past that of just your degree, requiring you to take multiple general education electives. I enjoyed all the classes I took, but there are two classes I wish to highlight the most in how they've impacted me and my learning.

The first class I want to reflect on was one I took back in my freshman year of college, and introduction to sociology class. This class changed how I see the world and how I think about issues that affect things socially. There are a lot of small details about life that are part of a bigger social dynamic that I never would have known about otherwise and now go into how I approach problems as an engineer. When presented with a problem, I'm more aware of and considerate of the social impacts it would have on us as a society, which goes into influencing me and my ethical decision making when considering the IEEE standards. You must consider how it could affect people more than just the initial appearance. I see engineering problems from more angles than previously before, something that wouldn't have been possible without taking this class.

Another class I want to highlight was a 3D modeling class I took in my junior year. This class was an expressive outlet for me, but it got me to think creatively about how I go about solving problems. I had to figure out how I wanted a certain shape to look or how to get the image I want to render in the position I wanted. I wanted a dense forest background, but that took a lot of time to render or was too much to handle as a whole, my solution was making only the front line of trees have leaves since that was all you could see, and making it so that backline trees were just trunks, and lastly placing a basic black background behind them all to give the illusion of a dense canopy and a deep forest. I apply this thinking when approaching problems as an engineer, if something isn't working, how can I approach it differently to see a new perspective to make it work in a way I never considered before.

General Education classes have broadened my horizons and expanded my problem-solving skills beyond the technical aspects of it. They helped me to gain new perspectives about problems, finding a solution from a different approach that's more creative, or they've shown me to consider the implications and affects they may have on individuals and society, being more mindful of the cultural impacts of what I am doing. I've thoroughly enjoyed the different general education electives, but more than just enjoying them I am glad I took them. I think a

broad range of general education electives are crucial to widening your understanding and problem-solving skills when it comes to approaching an issue, especially as an engineer. In such a technical major, it's important to not get swept up in the technical aspects of a problem and to consider the other dimensions of the problem, and general education electives do just that.